

Troy Mayer

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EDUCATION

Purdue University, School of Mechanical Engineering and School of Science

B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics

Clubs: Club Swimming, House Band

West Lafayette, IN

May 2028

GPA:3.4/4

PROFESSIONAL EXPERIENCE

Boston Dynamics

Sensors and Perception Co-Op – Atlas

Waltham, MA

July 2026 – Present

- Built a forward-kinematics pipeline that computes where each limb is and how fast it's moving at impact, propagating 6-DOF floating-base poses through the joint chains with Rodrigues' rotations and rigid-body velocity composition.
- Designed a delta-v impact-severity metric that stays accurate when sensors miss the sharpest part of an impact, validating the filtering against synthetic pulses of known size and measuring the metric's noise floor on quiet data.
- Built an impact forensics pipeline and Flask/Plotly dashboard on a Linux VM covering ~28k impact events from ~900 robot test logs, with ETL from a time-series telemetry store into SQLite, a 3D impact-location heatmap, and per-event waveform inspection used by engineers during testing.
- Modeled hardware thermal behavior in Python (Pandas, SciPy), using Savitzky-Golay filtering and multi-start least squares to fit heating curves and predict steady-state temperatures across hundreds of units without running each test to completion.

Caterpillar (via Purdue Data Mine)

Machine Learning Researcher

West Lafayette, IN

January 2026 – May 2026

- Built an LSTM model forecasting next-day primer, topcoat, and solvent consumption from five years of paint-line data, with Monte Carlo confidence intervals on each forecast, achieving a 4.7% reduction in material usage.
- Built Python pipelines to ingest, clean, and consolidate five years of noisy legacy data into one daily series, contributing to a 3.3% reduction in manufacturing cycle time.

Chortos Lab – Soft Adaptive Systems and Structures

Undergraduate Researcher

West Lafayette, IN

August 2025 – May 2026

- Built data acquisition systems in LabVIEW and MATLAB, processing and normalizing datasets of 50,000+ points to quantify sub-millimeter material behavior.
- Wrote a Python computer-vision tracker resolving cantilever displacement to 0.01 mm, guiding gel formulations that increased force output by over 50%.

PROJECTS

Purdue ASME – Biomechanics

Active Orthosis – Electronics Team Lead

West Lafayette, IN

April 2024 – Present

- Designed and implemented a 1 kHz PI velocity controller in C++ on an ESP32-S3, tuned from a first-order motor model with feed-forward and anti-windup, and protected by 10 fault monitors, a hardware watchdog, and a kill switch.
- Cut EMG-to-motion latency from 103 to 43 ms using a Bayesian EMG estimator, validated in closed-loop simulation on real EMG with 26 automated tests in CI.
- Led a 5-engineer team and redesigned the 4-channel EMG board in KiCad with an INA333 front end, right-leg drive, and ISO7763 motor isolation, raising cross-day gesture decoding from 31% to 87% in emulation.

Legged Robot State Estimation

Personal Project

August 2026 – Present

- Built a contact-aided error-state EKF in C++/Eigen fusing IMU and leg kinematics at 1 kHz, achieving 0.013 m/s velocity RMSE over a 60-second walk vs. 1.38 m/s for IMU-only dead reckoning.
- Verified filter Jacobians against finite differences in GoogleTest, ran each 1 kHz update in ~20 μ s, and showed roll, pitch, and velocity stay bounded while position and yaw drift.
- Stress-tested contact latency and gated contact detection on gait timing, removing 6.6° of yaw drift.

SKILLS & INTERESTS

- **Tools:** C++, Python, MATLAB, SQL, LabVIEW, CAD (NX, Fusion, Creo), FEA Analysis, KiCad, Snowflake, Git, Linux, Eigen, MuJoCo
- **Languages:** English (Native), Turkish (Fluent), Spanish (Fluent - New Jersey Seal of Bilinguality).
- **Hobbies:** Lifting, Hiking, Guitar, Saxophone